Lecture 4

Dr. Kiran Kumari

Department of Electrical Engineering
Indian Institute of Science
Bengaluru, India

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Sliding mode control and its application

Linearization of non-linear system:

1. First, find the equilibrium points
2. Then, linearize the system about each equilibrium point or desired equilibrium point.

For a second order system: $\dot{x} = f(x)$, $x \in \mathbb{R}^2, f(x) \in \mathbb{R}^2$

$$\dot{x}_1 = f_1(x_1, x_2)$$
$$\dot{x}_2 = f_2(x_1, x_2)$$

Linearization about $x_e = 0$ is obtained as

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{x=x_e}$$

Let $z = x - x_e$, $\dot{z} = \dot{x} - \dot{x}_e$.
This implies the linearized system is represented as $\dot{z} = Az$

For the system $\dot{x} = f(x, u)$ where $x \in \mathbb{R}^2$ and $u \in \mathbb{R}^2$, control matrix is obtained as

$$B = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \end{bmatrix}_{x=x_e, u=u_e}$$

The linearized system is $\dot{z} = Az + B(u - u_e)$.

- $\dot{x} = f(x, u)$, $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$
- $\dot{x} = 0$ gives n equations, so we need to have m approximations to solve n equations for $n + m$ variables.
- These assumptions can be on the control variable or on the state variable.

Vander Pal's Equation:

$$\ddot{y}(t) - \mu(1 - y^2(t))\dot{y}(t) + y(t) = 0, \quad \text{where} \quad \mu > 0$$

Inner product

Inner product of two vectors in $\mathbb{R}^n$

$$< x, y >= x^T y = \sum_{i=1}^{n} x_i y_i$$

Length of a vector $x \in \mathbb{R}^n$

It is a norm of a vector represented as $\|x\|$ which is a non-negative real number.

$$\|.\| : \mathbb{R}^n \rightarrow \mathbb{R}^+$$

Euclidian norm:

$$\|x\|_2 \triangleq \sqrt{x_1^2 + x_2^2 + ... + x_n^2}$$

Properties of norm:

1. Positivity: $\|x\| = 0$, if and only if $x = 0$
2. Scalar multiplication: $\|\alpha x\| = |\alpha|\|x\|$, where $\alpha \in \mathbb{R}$
3. Triangle inequality: $\forall x, y \in \mathbb{R}^n$, $\|x + y\| \leq \|x\| + \|y\|$

p^{th} norm of a vector:

$1 \leq p \leq \infty, x \in \mathbb{R}^n$

$$\|x\|_p = \left(|x_1|^p + |x_2|^p + ... + |x_n|^p\right)^{1/p}$$

Infinity norm:

$$\|x\|_\infty = \max_i \{|x_i|\}, \quad i \in 1, 2, ...n$$

Distance

The notion of distance between two vectors x and y can be represented as the norm

$$\begin{aligned} d(x, y) &= \|x - y\|_2 \\ &= \sqrt{(x - y)^T (x - y)} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2} \end{aligned}$$

1-norm:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

Cauchy Schwartz inequality

Let $x, y \in \mathbb{R}^n$ then

$$|< x, y >| \leq \|x\|\|y\|$$

The inner product is a scalar quantity, so we can take its absolute value, and this absolute value satisfies the inequality above. Also, equality holds when x and y are linearly dependent.

Some relationships between different norms

- $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$
- $\|x\|_1 \leq \sqrt{n}\|x\|_2 \leq n\|x\|_\infty$

Topology definitions

Open Ball

An open ball in $\mathbb{R}^n$ of radius $r > 0$ centered around a point x, denoted by

$$B(x, r) := \{y \in S \mid d(x, y) < r\}$$

where $S \subset \mathbb{R}^n$.

Open Set

A subset $A \subset \mathbb{R}^n$ is said to be open set if for all $x \in A$ there exists a $r_x > 0$ such that $B(x, r_x) \subset A$.

Interior point of A

Let $A \subset \mathbb{R}^n$, $x \in A$. Then x qualifies to be an interior point of A if there exists a ball $B(x, r)$ such that all points in $B(x, r)$ belongs to A.

Open Set

A set $A \subset \mathbb{R}^n$ is said to be open if all its points are interior points, $A = int(A)$.

Complement of a set

$A \subset \mathbb{R}^n$, the complement of A is $A^C = \mathbb{R}^n \setminus A$, where $\setminus$ is set minus symbol.

Adherent points/Closure points

Given a set A, $x \in \mathbb{R}^n$ is called an adherent point of A if every ball around 'x' contains at least one point from A. Formally, $\forall r > 0$, $B(x, r) \cap A \neq \emptyset$.

Bounded Set

A set 'A' is called bounded if it is contained in some ball, i.e., $A \subset B(x, r)$ for some $x \in \mathbb{R}^n$, $r > 0$.

Closed Set

The set 'A' is said to be closed if and only if it contains all its adherent points, i.e., $A = \overline{A}$, where $\overline{A}$ is set of Adherent points.
A closed set is also defined as a complement of the open set.

Limit points

A point $x \in \mathbb{R}^n$ is called a limit point of a set A if every neighborhood of 'x' contains a point of 'A' distinct from 'x'.

Closed Set

A set is said to be closed if it contains all its limit points.

Relation between adherent points and limit points

$$\overline{A} = A \cup A^{'}$$

$\overline{A}$ - closure of A, $A^{'}$- set of all limit points.

Observations

1. Every limit point is an adherent point, but the converse is not true.
2. An adherent point, which is not a limit point, is an isolated point.

Compact Set

A set 'A' is called compact if it is closed and bounded.

Boundary of a set

$$\partial A = \overline{A} \cap \overline{(\mathbb{R}^n \setminus A)}$$

- On a real line, closed intervals/sets are also compact.
- The only sets which are both open and closed are the set $\mathbb{R}^n$ and $\emptyset$.

Connnected set

A set is called disconnected if there exist two sets Y, Z such that $Y \cup Z = X$ and $\overline{Y} \cap Z = Y \cap \overline{Z} = \emptyset$.
A set is called connected if one cannot find a separation as given above, i.e., it is not disconnected.

Thank You